

Exercise 5

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- show that the entropy is

$$S(B, T) = Nk_B \left[\ln(2 \cosh x) - x \tanh x \right], \quad x = \frac{\mu B}{k_B T}.$$

- Show that S depends only on the ratio

$$x = \frac{\mu B}{k_B T}.$$

- Explain why, in an adiabatic process ($S = \text{const}$), this implies

$$\frac{B}{T} = \text{const}, \quad \text{hence} \quad T_f = T_i \frac{B_f}{B_i}.$$

Exercise 6

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- Evaluate the final temperature for

$$T_i = 1.2 \text{ K}, \quad B_i = 2.0 \text{ T}, \quad B_f = 2.0 \times 10^{-2} \text{ T}.$$

- Find the limiting forms of the entropy:

$$S(B, T) \quad \text{for} \quad x \ll 1 \quad \text{and} \quad x \gg 1.$$

- Interpret these two limits physically.